

TP1 Test Cases

Team 53 | CSE 360 | Phase One. Part 1 lists the input validation tests. They are automated in TP1InputValidationTesting.java, and the IDs match the IDs in that class. Each test compares the value the code returns with the expected result. Part 2 lists feature tests for account deletion and the related interface changes. They are run by hand in the application.

Part 1: Input Validation Tests

Username (UserNameRecognizer): 20/20 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
UN-01	abcd	Shortest valid UserName (4 characters)	Accept	Accept	PASS
UN-02	[a x 32]	Longest valid UserName (32 characters)	Accept	Accept	PASS
UN-03	Josh_99	Letters, a digit, and a separator are accepted	Accept	Accept	PASS
UN-04	a.b-c&d_e	Each of the four separators (. - & _) is accepted	Accept	Accept	PASS
UN-05	(empty string)	Empty input is rejected	Reject: The input is empty.	Reject: The input is empty.	PASS
UN-06	(null)	Null input is rejected without crashing	Reject: The input is empty.	Reject: The input is empty.	PASS
UN-07	abc	UserName one character under the minimum	Reject: A UserName must have at least 4 characters.	Reject: A UserName must have at least 4 characters.	PASS
UN-08	[a x 33]	UserName one character over the maximum	Reject: A UserName must have no more than 32 characters.	Reject: A UserName must have no more than 32 characters.	PASS
UN-09	9lives	First character must be a letter (digit rejected)	Reject: A UserName must start with A-Z or a-z.	Reject: A UserName must start with A-Z or a-z.	PASS
UN-10	_josh	First character must be a letter (separator rejected)	Reject: A UserName must start with A-Z or a-z.	Reject: A UserName must start with A-Z or a-z.	PASS
UN-11	josh__99	Two separators in a row are rejected	Reject: A separator must be followed by A-Z, a-z, or 0-9.	Reject: A separator must be followed by A-Z, a-z, or 0-9.	PASS
UN-12	josh_	A trailing separator is rejected	Reject: A separator must be followed by A-Z, a-z, or 0-9.	Reject: A separator must be followed by A-Z, a-z, or 0-9.	PASS
UN-13	josh smith	A space is not an allowed character	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS
UN-14	josh@asu	A disallowed symbol (@) is rejected	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS
UN-15	jo!	Illegal character found before the length check runs	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
UN-16	[a x 31]_	32 characters ending in a separator	Reject: A separator must be followed by A-Z, a-z, or 0-9.	Reject: A separator must be followed by A-Z, a-z, or 0-9.	PASS
UN-17	[a x 32]_	Separator as the 33rd character (too long)	Reject: A UserName must have no more than 32 characters.	Reject: A UserName must have no more than 32 characters.	PASS
UN-18	josé99	A non-ASCII letter is rejected	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	Reject: A UserName may only contain A-Z, a-z, 0-9, and the separators -, _, ., and &.	PASS
UN-19	(null)	Update Username page: null value is rejected before the recognizer runs	Reject: Enter a UserName.	Reject: Enter a UserName.	PASS
UN-20	josh_99	Update Username page: a valid value is passed through to the recognizer	Accept	Accept	PASS

Password (Model.evaluatePassword): 16/16 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
PW-01	Abcdef1!	Shortest valid password (8 characters, all four character types)	Accept	Accept	PASS
PW-02	Aa1! [x x 16]	Longest valid password (20 characters)	Accept	Accept	PASS
PW-03	(empty string)	Empty input is rejected	Reject: The password is empty!	Reject: The password is empty!	PASS
PW-04	Aa1! [x x 17]	One character over the maximum (21 characters)	Reject: The password may not exceed 20 characters!	Reject: The password may not exceed 20 characters!	PASS
PW-05	Abc1!	Too short: only the minimum-length rule fails	Reject: Long Enough; conditions were not satisfied	Reject: Long Enough; conditions were not satisfied	PASS
PW-06	Abcdefg1	Missing special character	Reject: Special character; conditions were not satisfied	Reject: Special character; conditions were not satisfied	PASS
PW-07	abcdefg1!	Missing uppercase letter	Reject: Upper case; conditions were not satisfied	Reject: Upper case; conditions were not satisfied	PASS
PW-08	ABCDEFG1!	Missing lowercase letter	Reject: Lower case; conditions were not satisfied	Reject: Lower case; conditions were not satisfied	PASS
PW-09	Abcdefgh!	Missing digit	Reject: Numeric digits; conditions were not satisfied	Reject: Numeric digits; conditions were not satisfied	PASS
PW-10	abcdefgh	Several requirements missing at once are all listed	Reject: Upper case; Numeric digits; Special character; conditions were not satisfied	Reject: Upper case; Numeric digits; Special character; conditions were not satisfied	PASS
PW-11	Abcdef 1!	A space is an invalid character (error index points at it)	Reject: An invalid character has been found!	Reject: An invalid character has been found!	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
PW-12	Abcdéf1!	A non-ASCII letter is an invalid character	Reject: An invalid character has been found!	Reject: An invalid character has been found!	PASS
PW-13	Aa1~`!@#\$%^&*()_-=+{	Special characters ~`!@#\$%^&*()_-=+{ are accepted	Accept	Accept	PASS
PW-14	Aa1}[] \:;'"<>,./	Special characters [] \:;'"<>,./ are accepted	Accept	Accept	PASS
PW-15	"Ab1 "	Invalid character reported ahead of the missing-requirement list	Reject: An invalid character has been found!	Reject: An invalid character has been found!	PASS
PW-16	(null)	Null input is rejected without crashing	Reject: The password is empty!	Reject: The password is empty!	PASS

Email address (EmailAddressRecognizer): 26/26 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
EM-01	josh@asu.edu	A simple valid address	Accept	Accept	PASS
EM-02	josh.l+asu@mail.asu.edu	Period and plus in the mailbox name; multi-part domain	Accept	Accept	PASS
EM-03	a@b.co	Shortest realistic valid address	Accept	Accept	PASS
EM-04	(empty string)	Empty input is rejected	Reject: The email address is empty!	Reject: The email address is empty!	PASS
EM-05	(null)	Null input is rejected without crashing	Reject: The email address is empty!	Reject: The email address is empty!	PASS
EM-06	[a x 64]@[b x 63].[b x 63].[c x 61]	Longest valid address (254 characters, 64-character mailbox name)	Accept	Accept	PASS
EM-07	[a x 64]@[b x 63].[b x 63].[c x 62]	One character over the maximum (255 characters)	Reject: The email address may not exceed 254 characters!	Reject: The email address may not exceed 254 characters!	PASS
EM-08	joshasu.edu	Missing the @	Reject: The email address must contain an "@"!	Reject: The email address must contain an "@"!	PASS
EM-09	josh@@asu.edu	More than one @	Reject: The email address may contain only one "@"!	Reject: The email address may contain only one "@"!	PASS
EM-10	@asu.edu	No mailbox name before the @	Reject: There must be a mailbox name before the "@"!	Reject: There must be a mailbox name before the "@"!	PASS
EM-11	[a x 65]@asu.edu	Mailbox name one over the 64-character limit	Reject: The part before the "@" may not exceed 64 characters!	Reject: The part before the "@" may not exceed 64 characters!	PASS
EM-12	jo sh@asu.edu	A space in the mailbox name	Reject: The character " " is not allowed before the "@"!	Reject: The character " " is not allowed before the "@"!	PASS
EM-13	.josh@asu.edu	Mailbox name starts with a period	Reject: The part before the "@" may not start or end with a period!	Reject: The part before the "@" may not start or end with a period!	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
EM-14	josh.@asu.edu	Mailbox name ends with a period	Reject: The part before the "@" may not start or end with a period!	Reject: The part before the "@" may not start or end with a period!	PASS
EM-15	jo..sh@asu.edu	Two periods in a row in the mailbox name	Reject: Two periods in a row are not allowed!	Reject: Two periods in a row are not allowed!	PASS
EM-16	josh@	No domain after the @	Reject: There must be a domain after the "@"!	Reject: There must be a domain after the "@"!	PASS
EM-17	josh@asu	Domain has no period	Reject: The domain must contain a period (e.g., asu.edu)!	Reject: The domain must contain a period (e.g., asu.edu)!	PASS
EM-18	josh@asu. .edu	Domain has an empty part	Reject: The domain has an empty part; check the periods!	Reject: The domain has an empty part; check the periods!	PASS
EM-19	josh@-asu.edu	Domain part starts with a hyphen	Reject: A domain part may not start or end with a hyphen!	Reject: A domain part may not start or end with a hyphen!	PASS
EM-20	josh@asu-.edu	Domain part ends with a hyphen	Reject: A domain part may not start or end with a hyphen!	Reject: A domain part may not start or end with a hyphen!	PASS
EM-21	josh@asu.e	Top-level domain is only one letter	Reject: The domain must end with at least two letters (e.g., .edu)!	Reject: The domain must end with at least two letters (e.g., .edu)!	PASS
EM-22	josh@asu.e1	Top-level domain contains a digit	Reject: The domain must end with letters only (e.g., .edu)!	Reject: The domain must end with letters only (e.g., .edu)!	PASS
EM-23	josh@as_u.edu	Illegal character in the domain	Reject: The character "_" is not allowed in the domain!	Reject: The character "_" is not allowed in the domain!	PASS
EM-24	josh@asu.edu.	Trailing period after the domain	Reject: The domain has an empty part; check the periods!	Reject: The domain has an empty part; check the periods!	PASS
EM-25	jösh@asu.edu	A non-ASCII letter in the mailbox name	Reject: The character "ö" is not allowed before the "@"!	Reject: The character "ö" is not allowed before the "@"!	PASS
EM-26	(null)	Update Email Address page: null value is rejected before the recognizer runs	Reject: Enter an email address.	Reject: Enter an email address.	PASS

Names: First, Middle, Last, Preferred First (InputValidation.checkName): 17/17 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
NM-01	Joshua	An ordinary name is accepted	Accept	Accept	PASS
NM-02	(empty string)	Empty value is accepted (clears the field)	Accept	Accept	PASS
NM-03	(null)	Null (dialog cancelled) is accepted and changes nothing	Accept	Accept	PASS
NM-04	Mary-Jane O'Neil	Hyphen, space, and apostrophe are accepted	Accept	Accept	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
NM-05	[a x 50]	Longest valid name (50 characters)	Accept	Accept	PASS
NM-06	[a x 51]	One character over the maximum (51 characters)	Reject: A name may not exceed 50 characters.	Reject: A name may not exceed 50 characters.	PASS
NM-07	Josh3	A digit is rejected	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	PASS
NM-08	Josh_	An underscore is rejected	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "_" is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "_" is not allowed.	PASS
NM-09	Jos.h	A period is rejected	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "." is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "." is not allowed.	PASS
NM-10	José	An accented letter is accepted	Accept	Accept	PASS
NM-11	" " (3 spaces)	Only spaces is rejected (no letters)	Reject: A name must contain at least one letter.	Reject: A name must contain at least one letter.	PASS
NM-12	-	Only a hyphen is rejected (no letters)	Reject: A name must contain at least one letter.	Reject: A name must contain at least one letter.	PASS
NM-13	[a x 51]3	Length check runs before the character check	Reject: A name may not exceed 50 characters.	Reject: A name may not exceed 50 characters.	PASS
NM-14	'	Only an apostrophe is rejected (no letters)	Reject: A name must contain at least one letter.	Reject: A name must contain at least one letter.	PASS
NM-15	O'Brien-Smith Jr	Letters mixed with apostrophe, hyphen, and space are accepted	Accept	Accept	PASS
NM-16	Zoë Ångström	Accented and other non-ASCII letters are accepted	Accept	Accept	PASS
NM-17	José3	A digit is still rejected after an accented letter	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	Reject: A name may contain only letters, spaces, hyphens, and apostrophes. The character "3" is not allowed.	PASS

Invitation role (InputValidation.validateRole): 9/9 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
RL-01	Admin	Admin is an accepted role	Accept	Accept	PASS
RL-02	Contributor	Contributor is an accepted role	Accept	Accept	PASS
RL-03	Viewer	Viewer is an accepted role	Accept	Accept	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
RL-04	Curator	Curator is an accepted role	Accept	Accept	PASS
RL-05	(null)	No role chosen (null)	Reject: Select a role for the invitation.	Reject: Select a role for the invitation.	PASS
RL-06	Select a role	The placeholder choice is not a role	Reject: Select a role for the invitation.	Reject: Select a role for the invitation.	PASS
RL-07	Superuser	A role that does not exist is rejected	Reject: The role "Superuser" is not one of the available roles.	Reject: The role "Superuser" is not one of the available roles.	PASS
RL-08	admin	Role names are case sensitive	Reject: The role "admin" is not one of the available roles.	Reject: The role "admin" is not one of the available roles.	PASS
RL-09	(empty string)	An empty role is rejected	Reject: The role "" is not one of the available roles.	Reject: The role "" is not one of the available roles.	PASS

Invitation code (InputValidation.validInvitationInput): 8/8 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
IC-01	ABC123	A typical code is accepted	Accept	Accept	PASS
IC-02	A	Shortest valid code (1 character)	Accept	Accept	PASS
IC-03	aaaaaaaaaa	Longest valid code (10 characters)	Accept	Accept	PASS
IC-04	aaaaaaaaaaa	One character over the maximum	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-05	(empty string)	Empty input is rejected	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-06	" " (3 spaces)	Whitespace-only input is rejected	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-07	(null)	Null input is rejected	Reject: enter a code of 1 to 10 characters	Reject: enter a code of 1 to 10 characters	PASS
IC-08	" ABC "	Spaces around a code are not trimmed; 5 characters is within the limit	Accept	Accept	PASS

Invitation deadline time (InputValidation.parseInvitationDeadline): 13/13 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/ Fail
TM-01	2026-10-01, 09:30	A valid time is combined with the date	Accept: 2026-10-01 09:30	Accept: 2026-10-01 09:30	PASS
TM-02	2026-10-01, 17:05	A valid afternoon time (24-hour clock)	Accept: 2026-10-01 17:05	Accept: 2026-10-01 17:05	PASS
TM-03	2026-10-01, 00:00	Lower boundary of the day	Accept: 2026-10-01 00:00	Accept: 2026-10-01 00:00	PASS
TM-04	2026-10-01, 23:59	Upper boundary of the day	Accept: 2026-10-01 23:59	Accept: 2026-10-01 23:59	PASS
TM-05	2026-10-01, " 09:30 "	Leading and trailing spaces are ignored	Accept: 2026-10-01 09:30	Accept: 2026-10-01 09:30	PASS

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
TM-06	2026-10-01, 9:30	Hour must have two digits	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-07	2026-10-01, 24:00	Hour 24 is not a valid time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-08	2026-10-01, 12:60	Minute 60 is not a valid time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-09	2026-10-01, ab:cd	Non-numeric time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-10	2026-10-01, (empty)	Empty time	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-11	2026-10-01, 09:30:00	Longer than 5 characters after trimming is rejected	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-12	(no date), 09:30	No date chosen	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS
TM-13	2026-10-01, (no time)	No time entered	Reject: not a valid time (returns null)	Reject: not a valid time (returns null)	PASS

Invitation deadline policy (InputValidation.validateInvitationDeadline): 6/6 test cases passed

#	Input	Rule Being Tested	Expected result	Actual result	Pass/Fail
DL-01	(cannot be parsed)	Deadline could not be parsed	Reject: Enter a valid expiration date and a time in the form HH:mm (24-hour clock).	Reject: Enter a valid expiration date and a time in the form HH:mm (24-hour clock).	PASS
DL-02	now + 1 day	One day from now is accepted	Accept	Accept	PASS
DL-03	now - 1 minute	One minute ago is rejected	Reject: The expiration date and time must be in the future.	Reject: The expiration date and time must be in the future.	PASS
DL-04	now + 30 days - 1 minute	Just inside the 30-day limit is accepted	Accept	Accept	PASS
DL-05	now + 30 days + 1 hour	Just past the 30-day limit is rejected	Reject: The expiration date and time must be within 30 days from now.	Reject: The expiration date and time must be within 30 days from now.	PASS
DL-06	now + 31 days	Well past the limit is rejected	Reject: The expiration date and time must be within 30 days from now.	Reject: The expiration date and time must be within 30 days from now.	PASS

Part 2: Feature Tests (account deletion and interface)

These cases cover account deletion and the related interface changes. Use disposable accounts for deletion checks. Each case lists the action and the expected result.

#	Test Case	Action	Expected result
FT-01	Cancel deletion	Select an account and choose No in the confirmation message.	The account remains available.
FT-02	Close the confirmation	Open the confirmation and press Escape or close the message.	The account is not deleted.

#	Test Case	Action	Expected result
FT-03	No account selected	Open the deletion page without selecting an account.	The Delete account button is disabled.
FT-04	Confirm deletion	Select another account and choose Yes.	The selected account is removed and the list is refreshed.
FT-05	Delete own account	Select the signed-in administrator account and confirm.	The app refuses to delete the account and explains why.
FT-06	Check administrator access	Try the deletion operation without an administrator session.	The operation is refused and no accounts are changed.
FT-07	Check deleted account access	After deletion, try the removed account's normal password and any active one-time password.	Neither gives access to the removed account.
FT-08	Check the selected account	Select an account and read the details and confirmation message.	The name, email and roles match the selection. The confirmation identifies the account to be deleted.
FT-09	Check an old selection	Remove an account, create another with the same username, and try deleting through the old selection.	The replacement account is not deleted.
FT-10	Check a failed database operation	Use a test database that prevents deletion and attempt to remove an account.	The operation reports failure and the account remains.
FT-11	Check changed permissions	Remove the signed-in user's administrator role before they attempt deletion.	Deletion is refused.
FT-12	Check renamed administrator	Change the signed-in administrator's username, then try to delete that account.	Self-deletion is still refused.
FT-13	Check invitation input	Try a blank invitation code and one longer than 10 characters.	The app asks for a code between 1 and 10 characters. A code within that length still needs to be a valid invitation.
FT-14	Check login input	Try a username longer than 32 characters or a password longer than 64 characters.	The login attempt is rejected.
FT-15	Check the confirmation buttons	Open the deletion confirmation and inspect the choices.	Yes is not the default action. No is the default and cancel action.
FT-16	Check the shared appearance	Open login and deletion and move between the application pages.	The shared style is applied, the fields are readable and the buttons remain usable.

Issues found by the input validation tests and fixed

Issue	Fix
Password check crashed on null input (PW-16)	Null is now rejected as an empty password.
Names of only spaces, a hyphen, or an apostrophe were accepted (NM-11, NM-12, NM-14)	A name must contain at least one letter.
Accented letters were rejected in names (NM-10, NM-16)	Any letter is now accepted.
The time 24:00 was accepted as midnight of the same day (TM-07)	The hour must be 00 to 23.
The invitation role defaulted to Admin (RL-05 to RL-09)	The drop-down starts on "Select a role", and a role must be chosen.
After a one-time password, the user was not sent back to Log In	The user now returns to the Log In page.
The name error alert always said "Name"	The alert now names the specific field.